S5 Glossary

Operational terms used in this reanalysis

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Terms

Meta-Noetic Pattern Space (MNPS). The exported feature chart used as input to downstream summaries. It is an operational representation of processed EEG, not a demonstrated latent neural state.

MNJ. Meta-Noetic Jacobian: the local derivative of the fitted chart flow. This analysis summarizes its trace, overall magnitude, antisymmetric rotation magnitude, and spectral radius per session.

Reachability. A local, model-based short-horizon description of variation reachable from a chart location. It is summarized here by covariance log-determinant, effective dimension, condition number, directional entropy, and a rotation proxy.

Subject anchoring. Coordinate normalization relative to a subject-level reference. It supports repeated-measure summaries but does not warrant interpreting absolute coordinate levels across children as a common scale.

ICC(A,1). Two-way absolute-agreement, single-measure intraclass correlation. It quantifies agreement while retaining a penalty for systematic session differences; it is not equivalent to Pearson correlation.

SEM and MDC95. Standard error of measurement and minimum detectable change at 95%. Both translate reliability estimates into the original endpoint scale.

Pseudo-replication. Artificially increasing apparent sample size by treating dependent epochs, windows, or repeated sessions as independent. The primary tests use one summary per child-session and one paired value per child.

BH-FDR. Benjamini–Hochberg false-discovery-rate adjustment applied to a stated endpoint family.

Exploratory analysis. A result-generating analysis whose raw p-values and effect sizes are used for hypothesis generation rather than confirmation. The balanced age contrasts, coordinate-position contrasts, 9D sensitivity analyses, and event-locked analyses are exploratory; the nine-endpoint dynamical family is reported with named BH-FDR sensitivities.